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*Logbook*

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| Session | Hours | Work Accomplished | Lessons Learnt/Design Notes | Goals for next session |
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| **26/3**<br>Onboarding session # 1 | 2 Hours | ➔ First onboarding session at Elizabeth Jolley<br>➔ Watched PAST Onboarding presentation<br>➔ Joined discord and explored the PAST GitHub page<br>➔ Researching projects that stand out and talking to PAST members to identify best match | Gained insight into the collaborative project environment within PAST. Learned that technical depth varies across departments and began leaning towards software-related projects due to prior exposure to python and major of software system engineering. | Finalise department choice and continue to review potential projects in software. |
| **27/3**<br><br>Eng. Pav | 2 Hours | ➔ Skill building with Arduino student kit:<br>➔ Completed traffic light project using green, yellow and red LED components and a breadboard circuit. Programmed light sequence for vehicles and pedestrians.<br>➔ Completed holiday lights project. Similar breadboard set up with additional lights and resistors programmed to flash like holiday/fairy lights. | Developed practical understanding of breadboard wiring and timing control using Arduino. Recognised the importance of resistor placement and loop timing in output behavior. | Explore input/output integration with sensors or potentiometers. |

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| **2/4**<br><br>Onboarding session # 2 | 2 Hours | ➔ Narrowed down department from mission, HR & software to software only<br>➔ Notified HR of my project selection via survey<br>➔ *Created logbook and selected project:* ***Machine Learning with ESP32 Camera***<br>➔ Created timeline spreadsheet for project<br>➔ Outlining project parameters and goals, introductory research<br>➔ Downloaded STM IDE and familiarized with set up for ESP32 Camera module programming | Decided to focus solely on software, recognizing the potential learning of combining camera data processing and machine learning on embedded systems.<br>Familiarised myself with the IDE that will be used to write and flash firmware to the microcontroller. | Start investigating star recognition methods and potential use of OpenCV |
| **3/4**<br><br>Eng. Pav | 3 Hours | ➔ Skill building with Arduino student kit:<br>➔ Completed musical keyboard project involving five push buttons which each represented different notes. Coded in Arduino IDE and wiring up breadboard with push buttons and resistors.<br>➔ Completed sports robot project involving making a servo move in synchronization with potentiometer dial. Used ratio of range of each component. Parts involved: servo, potentiometer, resistors, capacitor, 9V battery, breadboard etc.<br>➔ Researched Arduino kits to purchase for machine learning skills<br>➔ Purchased ESP32 camera module and ML kit | Strengthened analog and digital IO skills, understood how potentiometers can be used for variable control and how analog values are scaled in microcontroller logic. | Begin preliminary testing with ESP32-CAM and validate image streaming functionality. |

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| **9/4**<br><br>Onboarding session # 3 | 2 Hours | ➔ Created digital environment/workspace in MS notebook:<br>➔ Expanded brief<br>➔ Created question set to guide research:<br>- How can I access previous star tracking projects or satellite projects with ESP32 camera into space?<br>- What are the features of the ESP32 camera?<br>- Do I need to worry about lens?<br>- What is machine learning?<br>- Where does ai start and finish with machine learning for camera recognition?<br>- What kind of algorithm do I need to use?<br>- What data needs to be retained from a star?<br>- What kind of reference data does the module need to be pre-programmed e.g. star charts?<br>- Does it take constant images or it is triggered by the algorithm?<br>➔ Start to accumulate resources, research papers and reports of similar projects:<br>- ArduSat<br>- "Design and performance of an open-source star tracker algorithm on commercial off-the-shelf cameras and computers"<br>- Openstartracker.org<br>- Astrometry.net | Developed a structured inquiry framework to avoid aimless research. Began curating academic and open-source materials relevant to the project. | Collect working code examples for image processing with ESP32 and OpenCV |

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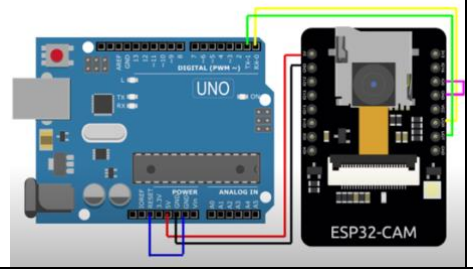
| | | - NASA astrophysics data system – database<br>- Introduction to C resources | | |
| **16/4**<br><br>Onboarding session #4 | 2 Hours | ➔ Received ESP32 Cam<br>➔ Downloaded ESP32 by Espressif systems onto Arduino IDE for playing around with Audino uno.<br>➔ Need to download OpenCV to raspberrypi<br>➔ Learn how to use STM32IDE with ESP32cam<br>➔ How to connect to Arduino or STM32 nucleo<br>➔ Use Openstartracker.org to play around – Downloaded from github<br>  | Refined understanding of the differences between classical vision algorithms and deep learning-based approaches. Noted tradeoffs between accuracy and processing overhead. | Start experimenting with image capture and transformation pipelines using ESP32 in a test set up. |

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| | | ➔ Star plot: Which map will I use: https://starplot.dev/<br><br>Resend link https://curtin-my.sharepoint.com/:w:/g/personal/14976441_student_curtin_edu_au/EZvCxlX9ZZpDt3HoUd-hMgsBDNaMVoTlK1GkoEEgk3-0cg?e=dxcyiQ<br><br>➔ Outline machine learning parameters – which kind? Deep machine learning, reinforcement learning, classic or non-deep learning.<br>- Phases of ML/Algorithm:<br>- Image classification<br>- Object detection<br>- Semantic Segmentation<br>- Pose estimation<br>- Hyperspectral tasks<br>- Compression techniques for complex programs (quantization and pruning)<br>- Edge deployment frameworks e.g. TensorFlow<br>- Lost in space mode????<br>➔ Go through software onboarding on GitHub – 5.01-5.04 to flesh out software/programming understanding<br>➔ Find ESP32 camera recognition project walk throughs to guide exploration with Arduino in | | |

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| | | python as preparation to try with C/C++<br>➔ e.g. https://www.youtube.com/watch?v=ujjg0LJ8hms<br>➔ and https://www.youtube.com/watch?v=zAY6hpb09i4<br>➔ Connect ESP32Camera and get it working with Audino uno at home to recognize objects using example sketch and tutorial.<br>➔ Planning to start putting basic research into proposal document, i.e. outline of components and functions | | |
| **23/4**<br><br>Onboarding session #5<br><br>Tuition free week | 3 hours | ➔ Esp32 camera linked to Arduino uno using the Arduino IDE to check camera is functioning and preview quality of image, get familiar with how to connect to microcontroller in a coding language I'm familiar with (Python) before attempting to use C++ | Realised importance of iterative hardware testing. Early validation avoids chasing software issues caused by physical malfunctions. | Set up STM32Cube IDE with appropriate tool chain and drive libraries for ESP32 communication. |
| **30/4**<br><br>Onboarding session #6 | 4 hours | ➔ Attempting to set up STM 32 IDE to recognize nucleo board and esp32 cam board.<br>- Using open CV to find algorithm | Learned how to link third-party OpenCV library to embedded toolchain. Clarified the expected data flow between sensor, algorithm and attitude estimation module. | Implement initial image processing script and test vector conversion using sample images. |

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| 6 Weeks mark | | - Collate research into report and fill out skeleton of report based on research and direction.<br>➔ Started draft of design concepts and overview document. | | |
| **1/5**<br><br>Home | 5 hours | ➔ Focused on refining research notes, integrating data sources and literature - organizing OneNote notebook for submission with logbook and report. Expanded draft report. Added a summary of possible attitude determination algorithms and embedded system limitations.<br>➔ Completed initial draft of Design Concepts and Overview document. Printed for proof reading and critique – Have I adequately addressed the brief thus far? | Emphasised structure documentation. Recognised the benefit of recording justifications and troubleshooting outcomes in real-time.<br><br>Recognised importance of synthesizing research information in Design Overview – some topics require far more research and exploration over the following stages of the project. | Complete a second draft of Design Concepts and Overview Document for Internal review<br><br>Flesh out holes in research and begin to test code in STM32 IDE<br><br>Link to OneNote Notebook: https://curtin-my.sharepoint.com/:o:/g/personal/14976441_student_curtin_edu_au/Ekbsgzez35BIqCBAWiDe6W4BgFa-L_CqCwQIRjfytBtg2g?e=jjlS6W |
| **7/5**<br><br>Onboarding session #7 | 3 hours | ➔ Review draft design concepts and overview and edit for cohesion and relevance<br>➔ Receive feedback on logbook and Design concept and overview from jack<br>➔ Need to go back and research space environment and what that means for fabrications/implement through report. | Getting feedback late impacts my whole workflow! Share work earlier on to have more input from team members with relevant expertise to save time and improve streamline of workflow. | |

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| **14/5**<br><br>Onboarding session #8<br><br>(Final session) | 6 hours | ➔ Documents for submission due Sunday 18th may:<br>➔ Logbook complete!<br>➔ Design Concept and Overview: complete!<br>➔ **OneNote Notebook:** https://curtin-my.sharepoint.com/:o:/g/personal/14976441_student_curtin_edu_au/Ekbsgzez35BIqCBAWiDe6W4BqFa-L_CqCwQIRjfytBtg2g?e=jjlS6W<br>➔ | Finalizing documentation early saved a lot of stress at end of semester while assignment due dates and tests are scheduled. | Get an interview! ☺ |

The logbook's purpose is to document the development of your skills, keep records of your project progress, and serve as a reference when creating your Design Concepts and Overview Document. We will be reviewing logbooks at the end of the Onboarding period so take pride in maintaining it!

Please feel free to customize your logbook however you like (changing table structure etc.), the most important thing is to document your learning! Note: readability should be a priority.

- *List any work accomplished and note down lessons learnt.*
- *Explain any design choices briefly (more in-depth explanation of your design decisions should go in your design concepts and overview document)*
- *Set any goals for next time.*
- *Include evidence and references (screenshots, photos, link*